

A smallest counterexample to the valency-variety inequality of Melnikov’s problem as stated on the Open Problem Garden

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Machine-generated note

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Abstract

The valency-variety $w(G)$ of a graph G is the number of distinct vertex degrees of G . Melnikov’s problem, as recorded in Jensen and Toft’s *Graph Coloring Problems* and transcribed on the Open Problem Garden, asks whether every graph G with at least two vertices satisfies $\chi(G) > \lceil \lfloor w(G)/2 \rfloor / (|V(G)| - w(G)) \rceil$. We exhibit a graph G_{37} on 37 vertices with $\chi(G_{37}) = 3$ and degree set exactly $\{0, 1, \dots, 29\}$, so that $w(G_{37}) = 30$ and the right-hand side equals $\lceil 15/7 \rceil = 3$: the inequality fails. A self-contained counting argument shows that no graph on at most 36 vertices violates the inequality, so 37 is the smallest possible order of a counterexample.

Caveat on interpretation. The counterexample is powered entirely by one isolated vertex z . The graph $G_{37} - z$ is a *connected* 36-vertex graph with $\chi = 3$ and degree set $\{1, \dots, 29\}$; it satisfies the inequality, and in fact meets the equivalent bound $|V| \leq (2\chi - 1)(|V| - w) + 1$ with equality. Adding z raises w from 29 to 30, hence $\lfloor w/2 \rfloor$ from 14 to 15, while leaving $|V| - w$ and χ unchanged. What is refuted is therefore the statement exactly as posed, with “any graph with at least two vertices” read literally. The version for connected graphs, or for graphs of minimum degree at least one, which is what most readers would call Melnikov’s problem, is untouched by this note and remains open; the referee’s mechanisation of our counting method shows that the method cannot decide that version once $|V| - w \geq 10$ (Remark 4.1).

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 5](#) records the full provenance.

1 Introduction

All graphs are finite and simple. For a graph G with vertex set $V(G)$ and $n = |V(G)|$, the *valency-variety* $w(G)$ is the number of distinct values taken by the degree function $d: V(G) \rightarrow \mathbb{Z}_{\geq 0}$; the degree 0 counts as a value. If $n \geq 2$ then two vertices share a degree (the n degrees lie in $\{0, \dots, n-1\}$ and 0 and $n-1$ cannot both occur), so $w(G) \leq n-1$ and

$$t(G) := n - w(G) \geq 1. \tag{1}$$

The Open Problem Garden [1] records the following problem, attributed to L. S. Melnikov, in the form below; the statement is quoted verbatim.

Question 1.1 (Melnikov’s valency-variety problem, [1]). The valency-variety $w(G)$ of a graph G is the number of different degrees in G . Is the chromatic number of any graph G with at least two vertices greater than

$$\left\lceil \frac{\lfloor w(G)/2 \rfloor}{|V(G)| - w(G)} \right\rceil ?$$

The accompanying discussion on [1] says that, according to Jensen and Toft [2, p. 90], the problem is due to Melnikov and was mentioned by Vizing [5] and Zykov [6]; that according to Zykov [6], Melnikov showed that the suggested lower bound would be best possible; and that a best possible *upper* bound on the chromatic number in terms of $|V(G)|$ and $w(G)$ is $\chi(G) \leq |V(G)| - \lfloor w(G)/2 \rfloor$, proved by Nettleton [4] and Dirac [3]. The restriction to graphs with at least two vertices is exactly the well-definedness condition (1): $n = 1$ is the only case with a zero denominator. The statement imposes no connectivity or minimum-degree hypothesis, and the adversarial referee confirmed (Appendix B) that the live Open Problem Garden page states the problem word for word as above and still lists it as open. We write

$$R(G) := \left\lceil \frac{\lfloor w(G)/2 \rfloor}{n - w(G)} \right\rceil$$

for the quantity in Question 1.1, so that the question asks whether $\chi(G) > R(G)$ for all G with $n \geq 2$.

Results. Section 2 gives a graph G_{37} on 37 vertices with $\chi(G_{37}) = 3 = R(G_{37})$, which answers Question 1.1 in the negative as it is stated. Section 3 proves that every graph on $2 \leq n \leq 36$ vertices satisfies $\chi > R$, so the order 37 is optimal. The proof of minimality is elementary but not short; it rests on a reformulation of the inequality as $n \leq (2\chi - 1)t + 1$ (Lemma 3.1), on a comparison between the sum of the degrees in the top half of the degree sequence and in the bottom half, and on a case analysis of four boundary configurations.

What this does and does not settle. We stress at the outset, and again in Remark 4.1, that the counterexample hinges on an isolated vertex, that $G_{37} - z$ satisfies the inequality, and that the connected (or minimum-degree-one) form of Melnikov’s question is not addressed. We could not read Jensen and Toft’s page 90 verbatim; if their wording, or that of Vizing or Zykov, restricts the problem to connected graphs, then this note refutes only the Open Problem Garden transcription (Remark 4.2). The literature search by the referee found no published resolution of Question 1.1 in either form, but the problem is little cited and the search is weak evidence of novelty (Remark 4.5).

2 The counterexample

2.1 The graph

The vertex set of G_{37} is the disjoint union

$$A = \{a_1, \dots, a_5\}, \quad B = \{b_0, \dots, b_5\}, \quad C = \{c_0, \dots, c_6\}, \quad T = \{t_1, \dots, t_6\}, \quad U = \{u_1, \dots, u_{12}\}, \quad \{z\},$$

of sizes $5 + 6 + 7 + 6 + 12 + 1 = 37$. The edge set is defined by the following rules, with no edges other than those listed.

- (E1) $A \cup B \cup C$ induces the complete tripartite graph with parts A, B, C .
- (E2) Every vertex of T is joined to every vertex of $B \cup C$.
- (E3) Each of $u_7, \dots, u_{12}$ is joined to every vertex of B .
- (E4) $N(u_7) \cap A = \{a_5\}$, $N(u_8) \cap A = \{a_4, a_5\}$, $N(u_9) \cap A = \{a_3, a_4, a_5\}$.
- (E5) For $i \in \{10, 11, 12\}$, u_i is joined to c_j exactly when $j \geq 13 - i$.
- (E6) For $1 \leq i \leq 6$, u_i is joined to b_j exactly when $i + j \geq 7$; in addition the six edges $u_1c_4, u_2c_5, u_3c_5, u_4c_6, u_5c_6, u_6c_6$ are present.

In particular $T \cup U$ is an independent set and z is isolated. Table 1 lists the complete neighbourhood of every vertex; it determines the graph and is the reference for all checks below. Vertices with identical neighbourhoods share a row.

Table 1: The graph G_{37} : every vertex with its degree and its full neighbourhood.

Vertex	Degree	Neighbourhood
z	0	$\emptyset$
u_1	1	$\{c_4\}$
u_2	2	$\{b_5, c_5\}$
u_3	3	$\{b_4, b_5, c_5\}$
u_4	4	$\{b_3, b_4, b_5, c_6\}$
u_5	5	$\{b_2, b_3, b_4, b_5, c_6\}$
u_6	6	$\{b_1, b_2, b_3, b_4, b_5, c_6\}$
u_7	7	$B \cup \{a_5\}$
u_8	8	$B \cup \{a_4, a_5\}$
u_9	9	$B \cup \{a_3, a_4, a_5\}$
u_{10}	10	$B \cup \{c_3, c_4, c_5, c_6\}$
u_{11}	11	$B \cup \{c_2, c_3, c_4, c_5, c_6\}$
u_{12}	12	$B \cup \{c_1, c_2, c_3, c_4, c_5, c_6\}$
a_1, a_2	13	$B \cup C$
$t_1, \dots, t_6$	13	$B \cup C$
a_3	14	$B \cup C \cup \{u_9\}$
a_4	15	$B \cup C \cup \{u_8, u_9\}$
a_5	16	$B \cup C \cup \{u_7, u_8, u_9\}$
c_0	17	$A \cup B \cup T$
c_1	18	$A \cup B \cup T \cup \{u_{12}\}$
c_2	19	$A \cup B \cup T \cup \{u_{11}, u_{12}\}$
c_3	20	$A \cup B \cup T \cup \{u_{10}, u_{11}, u_{12}\}$
c_4	21	$A \cup B \cup T \cup \{u_1, u_{10}, u_{11}, u_{12}\}$
c_5	22	$A \cup B \cup T \cup \{u_2, u_3, u_{10}, u_{11}, u_{12}\}$
c_6	23	$A \cup B \cup T \cup \{u_4, u_5, u_6, u_{10}, u_{11}, u_{12}\}$
b_0	24	$A \cup C \cup T \cup \{u_7, \dots, u_{12}\}$
b_1	25	$A \cup C \cup T \cup \{u_6, \dots, u_{12}\}$
b_2	26	$A \cup C \cup T \cup \{u_5, \dots, u_{12}\}$
b_3	27	$A \cup C \cup T \cup \{u_4, \dots, u_{12}\}$
b_4	28	$A \cup C \cup T \cup \{u_3, \dots, u_{12}\}$
b_5	29	$A \cup C \cup T \cup \{u_2, \dots, u_{12}\}$

2.2 Statement

Theorem 2.1 (Main theorem). *The graph G_{37} of Table 1 has 37 vertices and 263 edges, its degree set is exactly $\{0, 1, \dots, 29\}$, so $w(G_{37}) = 30$ and $t(G_{37}) = 7$, and $\chi(G_{37}) = 3$. Consequently*

$$R(G_{37}) = \left\lceil \frac{\lfloor 30/2 \rfloor}{37 - 30} \right\rceil = \left\lceil \frac{15}{7} \right\rceil = 3 = \chi(G_{37}),$$

and G_{37} does not satisfy $\chi(G_{37}) > R(G_{37})$.

Corollary 2.2. *The answer to Question 1.1, as stated, is negative.*

Proof. G_{37} has at least two vertices and $\chi(G_{37}) = 3 \not> 3 = R(G_{37})$ by Theorem 2.1. $\square$

Idea of proof. The degrees are read off Table 1: $|B \cup C| = 13$, $|A \cup B| = 11$, $|A \cup C| = 12$, $|T| = 6$, and the vertices of U have been attached so that u_i has degree i , c_j gains exactly j neighbours in U ,

and b_j gains exactly j neighbours among $u_1, \dots, u_6$ on top of the six from $u_7, \dots, u_{12}$. The three sets

$$I_1 = A \cup T \cup \{u_1, \dots, u_6, u_{10}, u_{11}, u_{12}, z\}, \quad I_2 = B, \quad I_3 = C \cup \{u_7, u_8, u_9\}$$

are independent and partition $V(G_{37})$, so $\chi \leq 3$, and $a_1 b_0 c_0$ is a triangle. The arithmetic $\lceil 15/7 \rceil = 3$ finishes. [\[full proof\]](#)

3 Minimality

Throughout this section G is a graph with $n \geq 2$ vertices, $k = \chi(G)$, $w = w(G)$ and $t = n - w \geq 1$. We call G *edgeless* if it has no edge (then $k = 1$), and we say that G *satisfies (R)* when the inequality below holds.

Lemma 3.1 (Reformulation). *For every graph G with $n \geq 2$,*

$$\chi(G) > R(G) \iff \lfloor w/2 \rfloor \leq (k-1)t \iff n \leq (2k-1)t + 1. \quad (R)$$

Idea of proof. For integers k and real x , $k > \lceil x \rceil$ iff $k-1 \geq x$; apply this to $x = \lfloor w/2 \rfloor / t$ and then eliminate the floor using $\lfloor w/2 \rfloor \leq N \iff w \leq 2N + 1$ and $w = n - t$. [\[full proof\]](#)

Theorem 3.2 (Minimality). *Every graph G with $2 \leq |V(G)| \leq 36$ satisfies $\chi(G) > R(G)$. Hence 37 is the smallest order of a counterexample to [Question 1.1](#), and G_{37} attains it.*

Idea of proof. By [Lemma 3.1](#) a counterexample violates (R). [Lemma 3.3](#) shows that graphs with $k \leq 2$ satisfy (R), so a counterexample has $k \geq 3$ and $n \geq (2k-1)t + 2 \geq 5t + 2$; if $n \leq 36$ this forces $t \leq 6$. [Proposition 3.7](#) shows that every graph with $t \leq 6$ satisfies (R). [\[full proof\]](#)

3.1 Graphs with at most two colours

Lemma 3.3 (Bipartite graphs). *Every edgeless graph and every bipartite graph with $n \geq 2$ satisfies (R). More precisely, a bipartite graph with at least one edge and without isolated vertices has $n \leq 3t$, and one with $r \geq 1$ isolated vertices has $n \leq 3t - 2r + 3 \leq 3t + 1$.*

Idea of proof. Without isolated vertices, fix a bipartition with parts of sizes $a \leq b$: all degrees lie in $\{1, \dots, b\}$, so $w \leq b$, and the degrees on the larger side lie in $\{1, \dots, a\}$ while the smaller side has only a vertices, so $w \leq 2a$; hence $n = a + b \geq \frac{3}{2}w$, i.e. $n \leq 3t$. Deleting $r \geq 1$ isolated vertices changes (n, w, t) to $(n-r, w-1, t-r+1)$. [\[full proof\]](#)

3.2 A degree-spread estimate

For the rest of the section G has even order $n = 2m$ and $k \geq 2$. List the degrees as $d_1 \leq d_2 \leq \dots \leq d_{2m}$, let L be a set of m vertices realising $d_1, \dots, d_m$ and H the remaining m vertices, and put

$$b = d_m, \quad \delta = d_1, \quad D = \sum_{v \in H} d(v) - \sum_{v \in L} d(v), \quad D_0 = \lfloor w^2/4 \rfloor.$$

Lemma 3.4 (Degree spread). *With the notation above:*

- (a) $D = \sum_{i=1}^{2m} |d_i - b| = 2e(H) - 2e(L)$; in particular D is even.
- (b) $D \geq D_0 = \lfloor w^2/4 \rfloor$.
- (c) If $w = 2r + 1$ and $b \leq \delta + r$, then $D \geq r(r+1) + (\delta + r - b)^2$.
- (d) If $w = 2r$ and $b \leq \delta + r - 1$, then $D \geq r^2 + (\delta + r - 1 - b)(\delta + r - b)$.

Idea of proof. (a): split $|d_i - b|$ according to $i \leq m$ or $i > m$, and count the edges inside and between L and H . (b): one occurrence of each of the w distinct degrees already contributes $\sum_i |x_i - b|$ over w distinct integers x_i , and at most two integers lie at any given positive distance from b , so this is at least $0 + 1 + 1 + 2 + 2 + \dots = \lfloor w^2/4 \rfloor$. (c), (d): all degrees are at least δ , so at most $b - \delta + 1$ of the chosen integers lie at or below b ; the minimum is attained by $\{\delta, \delta + 1, \dots, \delta + w - 1\}$ and evaluates to the stated expressions. [\[full proof\]](#)

3.3 Incorporating a colouring

Fix a proper k -colouring of G , let h_i be the number of vertices of colour i in H , and put

$$a = \min_i h_i, \quad e = e(L), \quad Q_k(m) = \min \left\{ \sum_{i=1}^k x_i^2 : x_i \in \mathbb{Z}_{\geq 0}, \sum_i x_i = m \right\}.$$

The minimum $Q_k(m)$ is attained exactly by the *balanced* vectors, those with $|x_i - x_j| \leq 1$ for all i, j ; moving one unit from an entry x to an entry $y \leq x - 2$ changes $\sum x_i^2$ by $-2(x - y - 1) < 0$. Set

$$X = D - D_0 \geq 0, \quad E = \sum_{i=1}^k h_i^2 - Q_k(m) \geq 0.$$

Lemma 3.5 (Colouring bounds). *With the notation above:*

- (a) $D + 2e + \sum_i h_i^2 \leq m^2$, equivalently $X + 2e + E \leq m^2 - D_0 - Q_k(m)$.
- (b) $b \leq m - a + e$.
- (c) E is even, and $E = 0$ forces (h_i) to be balanced, so that $a = \lfloor m/k \rfloor$.
- (d) $D_0 = m^2 - tm + \lfloor t^2/4 \rfloor$ and $\frac{m^2}{k} \leq Q_k(m) \leq tm - \lfloor t^2/4 \rfloor$.
- (e) For $t = 1, \dots, 6$ respectively, $m \leq k, 2k - 1, 3k - 1, 4k - 2, 5k - 2, 6k - 2$.

Idea of proof. (a): colour classes are independent, so $2e(H) \leq m^2 - \sum_i h_i^2$, and $2e(H) = D + 2e$ by Lemma 3.4(a). (b): the m -th vertex lies in L , misses the $\geq a$ vertices of its own colour in H , and has at most e neighbours inside L . (c): $x^2 \equiv x \pmod{2}$. (d): $w = 2m - t$ and Cauchy–Schwarz. (e): (d) is a quadratic inequality in m ; one checks that its left side minus its right side, at m one larger than each listed bound, equals $1 + \frac{1}{k}, 1, 2, \frac{1}{k}, 1 + \frac{1}{k}, 3 + \frac{1}{k}$ respectively, and that the quadratic is increasing from there on. [full proof]

Table 2: The four boundary configurations of Lemma 3.5(e) that need a separate argument. The “budget” is $m^2 - D_0 - Q_k(m) = tm - \lfloor t^2/4 \rfloor - Q_k(m)$, the right-hand side of Lemma 3.5(a).

t	m	w	$Q_k(m)$	budget	a when $E = 0$
1	k	$2(m - 1) + 1$	k	0	1
3	$3k - 1$	$2(m - 2) + 1$	$9k - 5$	0	2
5	$5k - 2$	$2(m - 3) + 1$	$25k - 18$	2	4
6	$6k - 2$	$2(m - 3)$	$36k - 22$	1	5

Lemma 3.6 (Boundary cases). *Let G have even order $2m$, $k \geq 2$ and $t \leq 6$.*

- (a) If $t \in \{1, 3, 5\}$ and G has no isolated vertex, then m is strictly smaller than the bound of Lemma 3.5(e): $m \leq k - 1, m \leq 3k - 2, m \leq 5k - 3$ respectively.
- (b) If $t = 6$ then $m \leq 6k - 3$, whether or not G has isolated vertices.

Idea of proof. Suppose m equals the bound. Table 2 gives the budget $m^2 - D_0 - Q_k(m)$, and Lemma 3.5(a) confines $(X, 2e, E)$ to it. For $t = 1, 3$ the budget is 0, so $X = e = E = 0$, a is as in the table, and Lemma 3.5(b) gives $b \leq m - a = r$; but $X = 0$ with $\delta \geq 1$ contradicts Lemma 3.4(c), which forces $b = \delta + r \geq r + 1$. For $t = 5$ the budget is 2; X is even because D and $D_0 = r(r + 1)$ are, $E \leq 2$ forces $a \geq 3$ by the transfer argument, and both $X = 0$ (then $b \leq m - 3 = r$) and $X = 2$ (then $b \leq m - 4$, penalty ≥ 4) contradict Lemma 3.4(c). For $t = 6$, $r = 6k - 5$ is odd, so $D_0 = r^2$ is odd and X is odd; the budget 1 forces $X = 1, e = E = 0, a = 5, b \leq r - 2$, and Lemma 3.4(d) with $\delta \geq 0$ gives $D \geq r^2 + 2$, i.e. $X \geq 2$. [full proof]

Proposition 3.7. *Every graph with $n \geq 2$ and $t \leq 6$ satisfies (R). More precisely, if G is not edgeless:*

- (a) *if n is even, then $n \leq (2k - 1)t$ for even t and $n \leq (2k - 1)t + 1$ for odd t , and in the latter case equality forces an isolated vertex;*
- (b) *if G has no isolated vertex, then $n \leq (2k - 1)t$;*
- (c) *if G has $r \geq 1$ isolated vertices, then $n \leq (2k - 1)t - (2k - 2)r + (2k - 1) \leq (2k - 1)t + 1$.*

Idea of proof. (a) is Lemma 3.5(e) together with Lemma 3.6, read as a statement about $n = 2m$. For (b) at odd order, add one isolated vertex: n and w both increase by one, t and k are unchanged, and (a) applies to the new graph. For (c), delete the isolated vertices and apply (b) to what remains, which has $t_0 = t - r + 1 \leq 6$ and the same chromatic number. [full proof]

4 Remarks

Remark 4.1 (What is refuted, and what is not). The whole mechanism of Theorem 2.1 is the isolated vertex z . The graph $G_{37} - z$ is connected (the referee verified this, and it is visible from Table 1: every vertex other than z has a neighbour in $B \cup C$, and $B \cup C$ is connected through A), has 36 vertices, degree set $\{1, \dots, 29\}$, so $w = 29$, $t = 7$, and $\chi = 3$; hence $R(G_{37} - z) = \lceil 14/7 \rceil = 2 < 3$ and $G_{37} - z$ satisfies the inequality of Question 1.1. It does so with no slack in the reformulation (R): $\lfloor w/2 \rfloor = 14 = (k - 1)t$ and $n = 36 = (2k - 1)t + 1$. Adding z raises w from 29 to 30 and hence $\lfloor w/2 \rfloor$ from 14 to 15, while t and χ stay fixed; this consumes exactly the unit of slack that the floor grants at odd w .

Consequently, a reader who understands “graph” in Question 1.1 as “connected graph” or “graph without isolated vertices” (a tacit convention in parts of the Russian school from which the problem comes) is not served by this note: for graphs of minimum degree at least one the inequality $\chi > R$ is, to our knowledge, still open, and Theorem 3.2 says nothing about how large a counterexample with minimum degree one would have to be. Worse for the method, the referee re-implemented the counting scheme of Section 3 (Lemmas 3.4 and 3.5 with the parity constraints on X and E) as a feasibility test in (X, e, E) and computed, for $k = 3$, the largest even order it does not exclude under the additional hypothesis $\delta \geq 1$: a counterexample with $\delta \geq 1$ is excluded for $t \leq 9$, but from $t = 10$ on it is not (at $t = 10$ a counterexample needs $n \geq 52$ and the method permits even orders up to 52; at $t = 11$ it needs 57 and the method permits 58; asymptotically the permitted order grows like $5.45t$ against the required $5t + 2$). The method of this note therefore provably cannot decide the minimum-degree-one version of Melnikov’s problem. The same mechanisation reproduces Proposition 3.7(a) exactly for $k = 2, \dots, 6$ and $t = 1, \dots, 7$ (for $k = 3$: maximal even orders 6, 10, 16, 20, 26, 30, 36), and shows that $G_{37} - z$ realises the largest even order the method permits at $k = 3$, $t = 7$, $\delta \geq 1$: the construction sits exactly on the boundary of the counting bound.

Remark 4.2 (Two residual risks). Neither the model nor the referee could read the primary source. Jensen and Toft [2, p. 90] is not freely available, and the papers of Dirac [3], Nettleton [4], Vizing [5] and Zykov [6] could not be obtained in full text. The mathematics of this note is self-contained and invokes no external theorem, so this does not affect the proofs; but it does affect what they refute. The referee fetched the live Open Problem Garden page and found that it reproduces the statement of Question 1.1 verbatim as quoted above, with “any graph G with at least two vertices” as the only hypothesis. If Jensen and Toft, or Vizing or Zykov, state the problem for connected graphs, the disproof evaporates (Remark 4.1).

Second, the placement of the ceiling is load-bearing. With $(n, w, \chi) = (37, 30, 3)$ the inequality fails only for the nesting as stated: $\chi > \lceil \lfloor w/2 \rfloor / t \rceil$ reads $3 > 3$, false; whereas $\chi > \lfloor w/2 \rfloor / t$ reads $3 > 2.14\dots$, true; $\chi > \lceil w/2 \rceil / t$ is true; and $\chi > \lfloor \lceil w/2 \rceil / t \rfloor$ reads $3 > 2$, true. Every other floor/ceiling nesting makes G_{37} a non-counterexample. The nesting used here is exactly the one on the live Open Problem Garden page and in the problem as posed to the model; a transcription slip at this point would void the whole note.

Remark 4.3 (The referee’s computational checks). The adversarial referee ([Appendix B](#); scripts in `verification_astra/scripts/melnikovs_valency_variety_problem/`, run with `python3` and `networkx 3.6.1`) rebuilt G_{37} from the rules (E1)–(E6) alone, not from the degree table, and found: 37 vertices, 263 edges, 603 triangles, components of sizes 36 and 1; degree sequence matching [Table 1](#) with zero mismatches, degree set exactly $\{0, 1, \dots, 29\}$, $w = 30$; I_1, I_2, I_3 of sizes 21, 6, 10 partitioning $V(G_{37})$ with 0 internal edges each; $\chi \geq 3$ established three independent ways (`networkx` bipartiteness test, exhaustive backtracking 2-colouring search failing while 3-colouring succeeds, and a hand-rolled BFS 2-colouring of the 36-vertex component forced onto the monochromatic edge u_3b_4); $t = 7$, $\lfloor w/2 \rfloor = 15$, $R = 3$; and for $G_{37} - z$: $n = 36$, $w = 29$, $t = 7$, $R = 2$, connected. For the minimality argument it re-derived by hand every displayed equation of the writeup (its equations (1)–(14), which correspond to [Lemmas 3.1](#) and [3.3](#) to [3.6](#) and [Proposition 3.7](#) here), checked the equivalence of [Lemma 3.1](#) exhaustively for all $2 \leq n \leq 199$ and all admissible w, k with 0 disagreements, verified by brute force that the six bounds of [Lemma 3.5\(e\)](#) are the exact integer maxima of the quadratic inequality for every $k \in \{2, \dots, 300\}$ and $t \in \{1, \dots, 6\}$, and re-implemented the whole counting scheme independently ([Remark 4.1](#)), corroborating that no counterexample exists below 37 vertices. It also checked all 1251 graphs on 2 to 7 vertices (the `networkx` atlas): every one satisfies the inequality, and the minimum slack $(2k-1)t+1-n$ in (R) is 0 at every order from 2 to 7, so the inequality is tight at every small order. The rewriter of this note independently rebuilt G_{37} from (E1)–(E6) and reproduced the edge count, the degree table, the three independent sets and the data for $G_{37} - z$ before typesetting [Table 1](#); as a sanity check, the degree sum from [Table 1](#) is $0 + 78 + 8 \cdot 13 + 45 + 140 + 159 = 526 = 2 \cdot 263$.

Remark 4.4 (Tightness and Melnikov’s claim). The observation that the slack in (R) is 0 at every order from 2 to 7 is consistent with Zykov’s report [6], relayed by [1], that Melnikov showed the suggested lower bound to be best possible, and it is an a priori indication that a counterexample was plausible: a bound that is tight at every small order has no room to absorb a degenerate perturbation such as the addition of an isolated vertex at odd w . The Dirac–Nettleton upper bound $\chi \leq n - \lfloor w/2 \rfloor$ [3, 4] is quoted only as context and is not used; for G_{37} it reads $3 \leq 22$.

Remark 4.5 (Novelty). The model stated that bibliographic novelty had not been checked. The referee’s four literature searches (counterexample, solved, valency-variety, Melnikov) returned nothing after 1995 on this specific problem, the Jensen–Toft problem archive has no update for this entry, and the Open Problem Garden page (last updated 3 March 2013) lists the problem as open with no comment or claimed resolution. This is weak evidence: the problem is rated of low importance and is rarely cited, and a reader of the 1960s with a tight example in hand could have made the same observation about isolated vertices. We cannot rule out that the counterexample is folklore. The minimality statement [Theorem 3.2](#) is, as far as we know, new.

5 Provenance

The mathematics in this note was produced without human intervention, in three stages.

(1) The construction and the minimality proof were found by the language model `gpt-6-astra` (reasoning effort *max*) in a single autonomous pass on 2026-09-16, as part of a sweep over the problems of the Open Problem Garden (catalog id `melnikovs_valency_variety_problem`, leg `attacks_opg`; original writeup in `attacks_opg/melnikovs_valency_variety_problem/output.md`). The model rated its claim “disproved” with high confidence, flagged that the counterexample has an isolated vertex “which the statement permits”, and stated that bibliographic novelty had not been checked.

(2) An independent adversarial referee agent (`claude-opus-5`) reviewed the writeup on 2026-09-17. It fetched the live Open Problem Garden entry and confirmed that the statement, the hypothesis “at least two vertices”, the discussion and the bibliography match the problem as posed to the model word for word; it rebuilt the graph from the prose alone and verified every claimed invariant ([Remark 4.3](#)); it re-derived all 26 steps of the writeup and labelled each `VALID`, finding no gap or error; it brute-forced the bounds of [Lemma 3.5\(e\)](#) for $k \leq 300$, checked all 1251 graphs on at most 7 vertices, and re-implemented

the counting scheme of [Section 3](#) independently, reproducing the writeup’s bounds and establishing the limitation recorded in [Remark 4.1](#). Its verdict was CONFIRMED, with the interpretation caveat that the disproof concerns the literal statement and not the minimum-degree-one version, and with the two residual risks of [Remark 4.2](#). Its report is reproduced verbatim in [Appendix B](#).

(3) On 2026-09-17 the writeup was rewritten into the present note by claude-fable-5-1. The text was reorganised with full proofs in [Appendix A](#). Deviations from the writeup, all declared here. The graph is presented as a complete neighbourhood table ([Table 1](#)); the writeup gave the six edge rules (E1)–(E6) and a degree table, and the neighbourhood table was generated by rebuilding the graph from the rules and checked against the referee’s data (edge count 263, degree sequence, independent sets, $G_{37} - z$). The edge count 263, the triangle count 603 and the connectedness of $G_{37} - z$ are the referee’s; the writeup did not state them. The writeup’s equations (1)–(14) appear as [Lemmas 3.1](#) and [3.3](#) to [3.6](#) and [Proposition 3.7](#). Three steps the writeup asserted were spelled out, in each case with an argument the referee had supplied or verified: that $\lfloor w^2/4 \rfloor$ is the least possible sum of distances from an integer to w distinct integers (counting integers by distance); that $\{\delta, \dots, \delta + w - 1\}$ minimises the sum of distances among w -sets of integers at least δ under the hypotheses of [Lemma 3.4\(c\),\(d\)](#) (the writeup said “the $2r + 1$ closest distinct integers at least δ are $\delta, \dots, \delta + 2r$ ”; the algebra with $s = b - \delta$ and $u = r - s$ is the referee’s); and that $E = 0$ pins down a , which uses the uniqueness of the balanced minimiser of $Q_k(m)$ (noted by the referee). In the $t = 5$ boundary the existence of a colour class of size at least 6 in H whenever some class has size at most 2 is made explicit. The edgeless case $k = 1$ is separated explicitly from the case $k \geq 2$ in [Lemma 3.3](#) and [Proposition 3.7](#), as the referee noted that [Lemma 3.5\(e\)](#) for $t = 6$ uses $k \geq 2$; the writeup handled edgeless graphs “directly” in one sentence. The statement of [Theorem 3.2](#) is the writeup’s “no counterexample has fewer than 37 vertices”. The content of [Remarks 4.1](#) to [4.5](#) is taken from the referee report and from the writeup’s own scope paragraph; the interpretation caveat was moved into the abstract at the rewriter’s initiative because the referee judged it the substantive reservation of the review. The bibliography is the one listed on [1]; none of the items [2, 3, 4, 5, 6] was read by the model, the referee or the rewriter. No mathematical content was changed, and nothing was found to be wrong.

No human mathematician has checked this argument. We circulate it for human review, not as a claim to a new resolution of Melnikov’s problem: it refutes the statement as transcribed on the Open Problem Garden, and the connected form of the question remains open.

A Full proofs

Theorem 2.1 (Main theorem). *The graph G_{37} of [Table 1](#) has 37 vertices and 263 edges, its degree set is exactly $\{0, 1, \dots, 29\}$, so $w(G_{37}) = 30$ and $t(G_{37}) = 7$, and $\chi(G_{37}) = 3$. Consequently*

$$R(G_{37}) = \left\lceil \frac{\lfloor 30/2 \rfloor}{37 - 30} \right\rceil = \left\lceil \frac{15}{7} \right\rceil = 3 = \chi(G_{37}),$$

and G_{37} does not satisfy $\chi(G_{37}) > R(G_{37})$.

Proof. Order and edges. The parts $A, B, C, T, U, \{z\}$ have sizes 5, 6, 7, 6, 12, 1, total 37. The degree column of [Table 1](#) sums to $0 + (1 + \dots + 12) + 8 \cdot 13 + (14 + 15 + 16) + (17 + \dots + 23) + (24 + \dots + 29) = 0 + 78 + 104 + 45 + 140 + 159 = 526$, so there are 263 edges.

Degrees. We verify [Table 1](#) against the rules (E1)–(E6); the degree of each vertex is the size of its listed neighbourhood. Note $|A| = 5$, $|B| = 6$, $|C| = 7$, $|T| = 6$, so $|B \cup C| = 13$, $|A \cup C| = 12$, $|A \cup B| = 11$.

- z is in no rule, so $d(z) = 0$.
- u_i , $1 \leq i \leq 6$: by (E6), $u_i \sim b_j$ iff $j \geq 7 - i$, i.e. $j \in \{7 - i, \dots, 5\}$, which has $i - 1$ elements; plus the one edge to C listed in (E6). So $d(u_i) = i$, with neighbourhood as tabulated.
- u_7, u_8, u_9 : all of B by (E3), plus 1, 2, 3 vertices of A by (E4); degrees 7, 8, 9.

- u_{10}, u_{11}, u_{12} : all of B by (E3), plus c_j for $j \geq 3, 2, 1$ respectively by (E5), i.e. 4, 5, 6 vertices of C ; degrees 10, 11, 12.
- a_1, a_2 : $B \cup C$ by (E1), nothing from U ((E4) names only a_3, a_4, a_5); degree 13.
- $t_1, \dots, t_6$: $B \cup C$ by (E2); degree 13.
- a_3, a_4, a_5 : $B \cup C$ by (E1), plus $\{u_9\}, \{u_8, u_9\}, \{u_7, u_8, u_9\}$ by (E4); degrees 14, 15, 16.
- c_j : $A \cup B$ by (E1) and T by (E2), that is 17 neighbours, plus its neighbours in U : from (E5), $u_i \sim c_j$ for $i \in \{10, 11, 12\}$ iff $i \geq 13 - j$, giving $\min(j, 3)$ vertices for $j \geq 0$ (none for $j = 0$, one for $j = 1$, two for $j = 2$, three for $j \geq 3$); from (E6), c_4 gets u_1 , c_5 gets u_2, u_3 , c_6 gets u_4, u_5, u_6 . Thus c_j has exactly j neighbours in U for every $0 \leq j \leq 6$, and $d(c_j) = 17 + j$.
- b_j : $A \cup C$ by (E1) and T by (E2), that is 18 neighbours, plus $u_7, \dots, u_{12}$ by (E3), plus from (E6) those u_i with $1 \leq i \leq 6$ and $i \geq 7 - j$, i.e. $i \in \{7 - j, \dots, 6\}$, which has j elements. So $d(b_j) = 18 + 6 + j = 24 + j$.

Hence the multiset of degrees is 0 (once), 1, ..., 12 (once each), 13 (eight times), 14, 15, 16 (once each), 17, ..., 23 (once each), 24, ..., 29 (once each): the degree set is $\{0, 1, \dots, 29\}$, $w(G_{37}) = 30$ and $t(G_{37}) = 37 - 30 = 7$.

Chromatic number. Let $I_1 = A \cup T \cup \{u_1, \dots, u_6, u_{10}, u_{11}, u_{12}, z\}$, $I_2 = B$, $I_3 = C \cup \{u_7, u_8, u_9\}$; these have sizes 21, 6, 10 and partition $V(G_{37})$. $I_2 = B$ is independent by inspection of Table 1 (no b_j lists a $b_{j'}$). I_3 : no c_j lists a $c_{j'}$ or any of u_7, u_8, u_9 , and u_7, u_8, u_9 list only vertices of $A \cup B$. I_1 : the vertices of A and T list only vertices of $B \cup C \cup \{u_7, u_8, u_9\}$; $u_1, \dots, u_6$ list only vertices of $B \cup C$; u_{10}, u_{11}, u_{12} list only vertices of $B \cup C$; z lists nothing. Since adjacency is symmetric, each I_i is independent, and $\chi(G_{37}) \leq 3$. The vertices a_1, b_0, c_0 are pairwise adjacent by (E1), so $\chi(G_{37}) \geq 3$. Hence $\chi(G_{37}) = 3$.

The inequality. $\lfloor w/2 \rfloor = 15$, $n - w = 7$, and $\lceil 15/7 \rceil = 3$ since $2 < 15/7 \leq 3$. Thus $R(G_{37}) = 3 = \chi(G_{37})$, and $\chi(G_{37}) > R(G_{37})$ is false. $\square$

Lemma 3.1 (Reformulation). *For every graph G with $n \geq 2$,*

$$\chi(G) > R(G) \iff \lfloor w/2 \rfloor \leq (k-1)t \iff n \leq (2k-1)t + 1. \quad (\text{R})$$

Proof. Put $x = \lfloor w/2 \rfloor / t$, a nonnegative rational. Since k is an integer, $k > \lceil x \rceil$ iff $k - 1 \geq \lceil x \rceil$, and since $k - 1$ is an integer, $\lceil x \rceil \leq k - 1$ iff $x \leq k - 1$. As $t > 0$, $x \leq k - 1$ iff $\lfloor w/2 \rfloor \leq (k - 1)t$. This proves the first equivalence. For the second, for an integer $N \geq 0$ we have $\lfloor w/2 \rfloor \leq N$ iff $w/2 < N + 1$ iff $w < 2N + 2$ iff $w \leq 2N + 1$; with $N = (k - 1)t$ and $w = n - t$ this reads $n - t \leq 2(k - 1)t + 1$, i.e. $n \leq (2k - 1)t + 1$. $\square$

Lemma 3.3 (Bipartite graphs). *Every edgeless graph and every bipartite graph with $n \geq 2$ satisfies (R). More precisely, a bipartite graph with at least one edge and without isolated vertices has $n \leq 3t$, and one with $r \geq 1$ isolated vertices has $n \leq 3t - 2r + 3 \leq 3t + 1$.*

Proof. If G is edgeless then $k = 1$, $w = 1$, $t = n - 1$ and (R) reads $n \leq (n - 1) + 1$, which holds with equality. Now let G be bipartite with at least one edge, so $k = 2$ and (R) reads $n \leq 3t + 1$.

No isolated vertices. Fix a bipartition $V(G) = P \cup R$ with $|P| = a \leq |R| = b$, $a + b = n$. Since there are edges and no isolated vertices, $a \geq 1$. A vertex of P has all its neighbours in R , so its degree lies in $\{1, \dots, b\}$; a vertex of R has degree in $\{1, \dots, a\} \subseteq \{1, \dots, b\}$. Thus all w distinct degrees lie in $\{1, \dots, b\}$ and $w \leq b$. Moreover the degrees occurring on R take at most a values, and the degrees occurring on P take at most $|P| = a$ values, so $w \leq 2a$. Hence $2n = 2a + 2b \geq w + 2w = 3w = 3(n - t)$, i.e. $n \leq 3t$.

$r \geq 1$ isolated vertices. Let G_0 be G with its isolated vertices deleted; G_0 is bipartite with at least one edge and no isolated vertex, with $n_0 = n - r \geq 2$ vertices. The degree 0 occurs in G and not in G_0 , and every other degree is unchanged, so $w_0 = w - 1$ and $t_0 = n_0 - w_0 = t - r + 1$. By the first part $n_0 \leq 3t_0$, hence $n = n_0 + r \leq 3(t - r + 1) + r = 3t - 2r + 3 \leq 3t + 1$ because $r \geq 1$. $\square$

Lemma 3.4 (Degree spread). *With the notation above:*

- (a) $D = \sum_{i=1}^{2m} |d_i - b| = 2e(H) - 2e(L)$; in particular D is even.
- (b) $D \geq D_0 = \lfloor w^2/4 \rfloor$.
- (c) If $w = 2r + 1$ and $b \leq \delta + r$, then $D \geq r(r + 1) + (\delta + r - b)^2$.
- (d) If $w = 2r$ and $b \leq \delta + r - 1$, then $D \geq r^2 + (\delta + r - 1 - b)(\delta + r - b)$.

Proof. (a) For $i \leq m$ we have $d_i \leq d_m = b$, and for $i > m$ we have $d_i \geq b$; so $\sum_{i=1}^{2m} |d_i - b| = \sum_{i \leq m} (b - d_i) + \sum_{i > m} (d_i - b) = mb - \sum_L d + \sum_H d - mb = D$. Writing $e(H, L)$ for the number of edges between H and L , $\sum_{v \in H} d(v) = 2e(H) + e(H, L)$ and $\sum_{v \in L} d(v) = 2e(L) + e(H, L)$, so $D = 2e(H) - 2e(L)$, which is even.

(b) Choose, for each of the w distinct degree values, one index i realising it; this gives w indices with distinct values $x_1 < \dots < x_w$, and by (a), $D \geq \sum_{j=1}^w |x_j - b|$. Order the x_j by increasing distance from b : exactly one integer lies at distance 0 from b , and at most two lie at each positive distance, so the j -th smallest of the distances $|x_j - b|$ is at least $\lceil (j-1)/2 \rceil$. Hence $\sum_j |x_j - b| \geq \sum_{j=1}^w \lceil (j-1)/2 \rceil = 0 + 1 + 1 + 2 + 2 + \dots$, which equals $r(r + 1)$ if $w = 2r + 1$ and r^2 if $w = 2r$; both equal $\lfloor w^2/4 \rfloor$.

(c) and (d). Every degree is at least $\delta = d_1$, and b is itself a degree, so $s := b - \delta \geq 0$. Let $S = \{x_1, \dots, x_w\}$ be the set of distinct degree values as in (b); all elements of S are integers $\geq \delta$. Let $p = |S \cap (-\infty, b]|$ and $q = w - p = |S \cap (b, \infty)|$. The elements of S at or below b are distinct integers in $[\delta, b]$, so $p \leq s + 1$, and their distances to b are distinct integers in $\{0, \dots, s\}$, so they sum to at least $0 + 1 + \dots + (p-1) = \binom{p}{2}$. The elements above b have distinct positive distances, summing to at least $1 + \dots + q = \binom{q+1}{2}$. Thus

$$D \geq f(p) := \binom{p}{2} + \binom{w-p+1}{2}, \quad f(p+1) - f(p) = p - (w-p) = 2p - w.$$

In case (c), $w = 2r + 1$ and the hypothesis $b \leq \delta + r$ says $s \leq r$, so $p \leq s + 1 \leq r + 1$; for $p \leq r$ we have $2p - w \leq -1 < 0$, so f is decreasing on $\{0, \dots, r + 1\}$ and $D \geq f(s + 1)$. Writing $u = r - s \geq 0$,

$$f(s+1) = \frac{s(s+1)}{2} + \frac{(2r-s)(2r-s+1)}{2} = \frac{(r-u)(r-u+1) + (r+u)(r+u+1)}{2} = r^2 + r + u^2,$$

which is $r(r + 1) + (\delta + r - b)^2$. In case (d), $w = 2r$ and the hypothesis $b \leq \delta + r - 1$ says $s \leq r - 1$, so $p \leq s + 1 \leq r$; for $p \leq r - 1$ we have $2p - w \leq -2 < 0$, so again $D \geq f(s + 1)$. Writing $u = r - 1 - s \geq 0$,

$$f(s+1) = \frac{s(s+1)}{2} + \frac{(2r-1-s)(2r-s)}{2} = \frac{(r-1-u)(r-u) + (r+u)(r+u+1)}{2} = r^2 + u^2 + u,$$

which is $r^2 + (\delta + r - 1 - b)(\delta + r - b)$. □

Lemma 3.5 (Colouring bounds). *With the notation above:*

- (a) $D + 2e + \sum_i h_i^2 \leq m^2$, equivalently $X + 2e + E \leq m^2 - D_0 - Q_k(m)$.
- (b) $b \leq m - a + e$.
- (c) E is even, and $E = 0$ forces (h_i) to be balanced, so that $a = \lfloor m/k \rfloor$.
- (d) $D_0 = m^2 - tm + \lfloor t^2/4 \rfloor$ and $\frac{m^2}{k} \leq Q_k(m) \leq tm - \lfloor t^2/4 \rfloor$.
- (e) For $t = 1, \dots, 6$ respectively, $m \leq k$, $2k - 1$, $3k - 1$, $4k - 2$, $5k - 2$, $6k - 2$.

Proof. (a) Every edge inside H joins two vertices of different colours, so $e(H) \leq \sum_{i < j} h_i h_j = \frac{1}{2}(m^2 - \sum_i h_i^2)$, using $\sum_i h_i = m$. By Lemma 3.4(a), $2e(H) = D + 2e(L) = D + 2e$, so $D + 2e + \sum_i h_i^2 \leq m^2$. Subtracting $D_0 + Q_k(m)$ from both sides gives $X + 2e + E \leq m^2 - D_0 - Q_k(m)$.

(b) Let v be a vertex of L with $d(v) = d_m = b$ and let c be its colour. The $h_c \geq a$ vertices of colour c in H are not adjacent to v , so v has at most $m - a$ neighbours in H . Each neighbour of v in L contributes a distinct edge inside L , so v has at most e neighbours in L . Hence $b \leq m - a + e$.

(c) Since $x^2 \equiv x \pmod{2}$ for every integer x , $\sum_i h_i^2 \equiv \sum_i h_i = m$ and likewise $Q_k(m) = \sum_i x_i^2 \equiv m$ for a minimising vector; so E is even. If $E = 0$ then (h_i) attains the minimum $Q_k(m)$; by the transfer argument preceding the lemma, a vector attaining the minimum has all entries within one of each other, so its entries are $\lfloor m/k \rfloor$ and $\lceil m/k \rceil$ and $a = \min_i h_i = \lfloor m/k \rfloor$.

(d) From $w = 2m - t$, $w^2/4 = m^2 - mt + t^2/4$, and $m^2 - mt$ is an integer, so $D_0 = \lfloor w^2/4 \rfloor = m^2 - tm + \lfloor t^2/4 \rfloor$. By Cauchy-Schwarz, $\sum_i x_i^2 \geq (\sum_i x_i)^2/k = m^2/k$ for every admissible vector, so $Q_k(m) \geq m^2/k$. By Lemma 3.4(b) and (a), $D_0 \leq D \leq m^2 - \sum_i h_i^2 \leq m^2 - Q_k(m)$, so $Q_k(m) \leq m^2 - D_0 = tm - \lfloor t^2/4 \rfloor$.

(e) By (d), $p(m) := m^2 - ktm + k\lfloor t^2/4 \rfloor \leq 0$. The quadratic p is increasing for $m \geq kt/2$. For each $t \leq 6$ let M_t be the claimed bound; then $M_t + 1 \geq kt/2$ (indeed $M_t + 1 > kt/2$ for $k \geq 2$ in each case), and we compute $p(M_t + 1)$:

$$\begin{aligned} t = 1 : & \quad p(k+1) = (k+1)^2 - k(k+1) = k+1 > 0; \\ t = 2 : & \quad p(2k) = 4k^2 - 4k^2 + k = k > 0; \\ t = 3 : & \quad p(3k) = 9k^2 - 9k^2 + 2k = 2k > 0; \\ t = 4 : & \quad p(4k-1) = (4k-1)^2 - 4k(4k-1) + 4k = 1 > 0; \\ t = 5 : & \quad p(5k-1) = (5k-1)^2 - 5k(5k-1) + 6k = k+1 > 0; \\ t = 6 : & \quad p(6k-1) = (6k-1)^2 - 6k(6k-1) + 9k = 3k+1 > 0. \end{aligned}$$

Dividing by k gives the residuals $1 + \frac{1}{k}$, 1 , 2 , $\frac{1}{k}$, $1 + \frac{1}{k}$, $3 + \frac{1}{k}$ quoted in the proof idea. Since p is increasing from $M_t + 1$ onwards, $p(m) > 0$ for all $m \geq M_t + 1$, so $p(m) \leq 0$ forces $m \leq M_t$. (For $t = 6$ and $m = M_6 = 6k - 2$ one has $p(6k - 2) = 4 - 3k \leq 0$ precisely because $k \geq 2$; this is the only place where $k \geq 2$ is needed, and it is where the bound is attained.) $\square$

Lemma 3.6 (Boundary cases). *Let G have even order $2m$, $k \geq 2$ and $t \leq 6$.*

- (a) *If $t \in \{1, 3, 5\}$ and G has no isolated vertex, then m is strictly smaller than the bound of Lemma 3.5(e): $m \leq k - 1$, $m \leq 3k - 2$, $m \leq 5k - 3$ respectively.*
- (b) *If $t = 6$ then $m \leq 6k - 3$, whether or not G has isolated vertices.*

Proof. In each case suppose, for a contradiction, that m equals the bound M_t of Lemma 3.5(e). We first verify Table 2. $Q_k(m)$ is the sum of squares of the balanced vector: for $m = k$ it is $k \cdot 1^2 = k$; for $m = 3k - 1$ it is $(k-1) \cdot 3^2 + 2^2 = 9k - 5$; for $m = 5k - 2$ it is $(k-2) \cdot 5^2 + 2 \cdot 4^2 = 25k - 18$; for $m = 6k - 2$ it is $(k-2) \cdot 6^2 + 2 \cdot 5^2 = 36k - 22$ (for $k = 2$ the balanced vectors are $(1, 1)$, $(3, 2)$, $(4, 4)$, $(5, 5)$, and the formulas still hold). The budget $tm - \lfloor t^2/4 \rfloor - Q_k(m)$ is then $k - 0 - k = 0$; $(9k - 3) - 2 - (9k - 5) = 0$; $(25k - 10) - 6 - (25k - 18) = 2$; $(36k - 12) - 9 - (36k - 22) = 1$. When $E = 0$, Lemma 3.5(c) gives $a = \lfloor m/k \rfloor = 1, 2, 4, 5$ respectively. Finally $w = 2m - t$ gives $r = m - 1$, $m - 2$, $m - 3$, $m - 3$ in the four rows, with $w = 2r + 1$ in the first three and $w = 2r$ in the last.

(a), $t = 1$ and $t = 3$. Here $\delta \geq 1$ because there is no isolated vertex. The budget is 0, so Lemma 3.5(a) gives $X = e = E = 0$. Then $a = 1$ (resp. 2) and Lemma 3.5(b) gives $b \leq m - a = r$. In particular $b \leq r < \delta + r$, so the hypothesis of Lemma 3.4(c) holds, and $D \geq r(r+1) + (\delta + r - b)^2 \geq r(r+1) + 1 = D_0 + 1$, i.e. $X \geq 1$, contradicting $X = 0$.

(a), $t = 5$. Again $\delta \geq 1$; $m = 5k - 2$, $r = m - 3$, $w = 2r + 1$, and the budget is 2: $X + 2e + E \leq 2$. Since D is even (Lemma 3.4(a)) and $D_0 = r(r+1)$ is even, X is even; so $X \in \{0, 2\}$.

We claim $E \leq 2$ forces $a \geq 3$. Suppose some $h_i \leq 2$. The other $k - 1$ entries sum to at least $5k - 2 - 2 = 5k - 4$; if they were all at most 5 their sum would be at most $5k - 5$, so some $h_j \geq 6$. Moving one unit from h_j to h_i changes $\sum h_i^2$ by $-2(h_j - h_i - 1) \leq -2(6 - 2 - 1) = -6$ and yields another admissible vector, so $\sum_i h_i^2 \geq Q_k(m) + 6$, i.e. $E \geq 6 > 2$. Hence $a \geq 3$.

If $X = 0$, then $2e + E \leq 2$ with E even: either $E = 0$, whence $a = 4$ and $e \leq 1$, so $b \leq m - a + e \leq m - 3$; or $E = 2$, whence $e = 0$ and $a \geq 3$, so $b \leq m - 3 = r < \delta + r$, and [Lemma 3.4\(c\)](#) gives $D \geq r(r + 1) + (\delta + r - b)^2 \geq D_0 + 1$, so $X \geq 1$: contradiction. If $X = 2$, then $e = E = 0$, so $a = 4$ and $b \leq m - 4 = r - 1$; hence $\delta + r - b \geq 1 + 1 = 2$ and [Lemma 3.4\(c\)](#) gives $D \geq r(r + 1) + 4$, i.e. $X \geq 4$: contradiction.

(b), $t = 6$. No assumption on δ is made, only $\delta \geq 0$. Here $m = 6k - 2$, $r = m - 3 = 6k - 5$ is odd, $w = 2r$, $D_0 = r^2$ is odd, and D is even; so $X = D - D_0$ is odd, in particular $X \geq 1$. The budget is 1, so $X = 1$ and $e = E = 0$. Then $a = 5$ and $b \leq m - 5 = r - 2 \leq \delta + r - 1$, so [Lemma 3.4\(d\)](#) applies: $D \geq r^2 + (\delta + r - 1 - b)(\delta + r - b) \geq r^2 + 1 \cdot 2 = D_0 + 2$, i.e. $X \geq 2$: contradiction. Hence $m \leq 6k - 3$. $\square$

Proposition 3.7. *Every graph with $n \geq 2$ and $t \leq 6$ satisfies (R). More precisely, if G is not edgeless:*

- (a) *if n is even, then $n \leq (2k - 1)t$ for even t and $n \leq (2k - 1)t + 1$ for odd t , and in the latter case equality forces an isolated vertex;*
- (b) *if G has no isolated vertex, then $n \leq (2k - 1)t$;*
- (c) *if G has $r \geq 1$ isolated vertices, then $n \leq (2k - 1)t - (2k - 2)r + (2k - 1) \leq (2k - 1)t + 1$.*

Proof. Edgeless graphs satisfy (R) by [Lemma 3.3](#), so assume G has an edge; then $k \geq 2$.

(a) Let $n = 2m$. By [Lemma 3.5\(e\)](#), $m \leq M_t$ with $M_t = k, 2k - 1, 3k - 1, 4k - 2, 5k - 2, 6k - 2$ for $t = 1, \dots, 6$. For $t = 2$ and $t = 4$ this gives $n \leq 4k - 2 = (2k - 1) \cdot 2$ and $n \leq 8k - 4 = (2k - 1) \cdot 4$. For $t = 6$, [Lemma 3.6\(b\)](#) improves the bound to $m \leq 6k - 3$, so $n \leq 12k - 6 = (2k - 1) \cdot 6$. For $t = 1, 3, 5$ we get $n \leq 2k = (2k - 1) \cdot 1 + 1$, $n \leq 6k - 2 = (2k - 1) \cdot 3 + 1$, $n \leq 10k - 4 = (2k - 1) \cdot 5 + 1$; and if equality holds then $m = M_t$, which by [Lemma 3.6\(a\)](#) is impossible without an isolated vertex.

(b) Let G have no isolated vertex; then $\delta \geq 1$ and $k \geq 2$. If n is even, (a) gives $n \leq (2k - 1)t$ directly for even t , and for odd t it gives $n \leq (2k - 1)t + 1$ with equality excluded, hence $n \leq (2k - 1)t$. If n is odd, let $G' = G + z$ be obtained by adding one isolated vertex. Then $n' = n + 1$ is even, $\chi(G') = k$, and since 0 was not a degree of G but is one of G' while all other degrees are unchanged, $w' = w + 1$ and $t' = n' - w' = t$. Applying (a) to G' : if t is even, $n + 1 \leq (2k - 1)t$, so $n \leq (2k - 1)t - 1$; if t is odd, $n + 1 \leq (2k - 1)t + 1$, so $n \leq (2k - 1)t$. In all cases $n \leq (2k - 1)t$.

(c) Let G have $r \geq 1$ isolated vertices and let G_0 be G with them deleted. Then G_0 has an edge, no isolated vertex, $n_0 = n - r \geq 2$ vertices, $\chi(G_0) = k$, $w_0 = w - 1$ and $t_0 = n_0 - w_0 = t - r + 1 \leq t \leq 6$. By (b), $n - r \leq (2k - 1)(t - r + 1)$, so

$$\begin{aligned} n &\leq (2k - 1)t - (2k - 1)r + (2k - 1) + r \\ &= (2k - 1)t - (2k - 2)r + (2k - 1) \\ &\leq (2k - 1)t - (2k - 2) + (2k - 1) = (2k - 1)t + 1, \end{aligned}$$

using $r \geq 1$ and $2k - 2 \geq 0$.

In every case $n \leq (2k - 1)t + 1$, which is (R). $\square$

Theorem 3.2 (Minimality). *Every graph G with $2 \leq |V(G)| \leq 36$ satisfies $\chi(G) > R(G)$. Hence 37 is the smallest order of a counterexample to [Question 1.1](#), and G_{37} attains it.*

Proof. Let G be a graph with $2 \leq n \leq 36$ and suppose $\chi(G) \leq R(G)$. By [Lemma 3.1](#), $n \geq (2k - 1)t + 2$. By [Lemma 3.3](#), G is neither edgeless nor bipartite, so $k \geq 3$ and $n \geq 5t + 2$. As $n \leq 36$, $5t \leq 34$, so $t \leq 6$. But then [Proposition 3.7](#) gives $n \leq (2k - 1)t + 1$, a contradiction. Hence every graph on 2 to 36 vertices satisfies $\chi > R$, and by [Theorem 2.1](#) the graph G_{37} on 37 vertices does not. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its step and equation numbers refer to that writeup, not to this note. The scripts it mentions are in `verification_astra/scripts/melnikovs_valency_variety_problem/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The writeup formalises the problem as: for every simple graph G with $|V(G)| \geq 2$,

$$\chi(G) > \lceil \text{floor}(w(G)/2) / (|V(G)| - w(G)) \rceil$$

where $w(G)$ is the number of distinct degrees, **with degree 0 counted as a degree** and with no connectivity or minimum-degree restriction. I fetched the live OpenProblemGarden entry (http://www.openproblemgarden.org/op/melnikovs_valency_variety_problem) and it reproduces exactly the statement in `prompt.md`, verbatim, with the sole restriction “any graph G with at least two vertices” and no further hypothesis.

Two observations argue that this literal reading is the intended one:

- The restriction $|V(G)| \geq 2$ is *precisely* the well-definedness condition: for $n \geq 2$ two vertices must share a degree, so $w \leq n-1$ and the denominator $n - w$ is positive; only $n = 1$ makes it zero. So the author did consider the degenerate cases and excluded exactly the one that breaks the formula — and did **not** exclude isolated vertices or disconnected graphs.
- Degree 0 counting toward w is forced by the definition “the number of different degrees”.

Two observations argue the other way, and they are the substantive caveat of this review:

- The counterexample is *entirely* powered by the isolated vertex. Deleting it leaves a connected 36-vertex graph that satisfies the inequality (I verified: $n=36, w=29, t=7, \text{RHS} = \lceil 14/7 \rceil = 2 < 3$). The writeup says so itself, honestly, in its final paragraph and in the caveats field.
- The primary source (Jensen–Toft, *Graph Coloring Problems*, p. 90, §4.13/4.14) could not be obtained verbatim. OPG is a secondary transcription. If Jensen–Toft (or Zykov/Vizing, from the Russian school where “graph” often tacitly means connected graph) state the problem for connected graphs, the disproof evaporates. This is the one load-bearing item I could not verify.

Would a reasonable author consider the problem resolved? Partly. The disproof is correct and genuinely non-trivial for the statement as posed and as recorded by OPG — the hard part (a connected 36-vertex 3-chromatic graph with 29 distinct degrees, which sits *exactly* on the boundary of the best available counting bound) is real extremal combinatorics, not a semantic trick. But the mathematical substance of Melnikov’s question survives: for graphs of minimum degree ≥ 1 the inequality is still open, and my mechanisation of the writeup’s own method (below) shows that method cannot decide it. A referee would classify this as “the stated conjecture is false; the interesting conjecture is still open”, not as “problem solved”. The writeup does not oversell this — its scope paragraph states the limitation explicitly — so I do not count it as interpretation gaming.

B.2 Step-by-step findings

Notation as in the writeup: $n = |V|$, $k = \chi$, $t = n - w$, $m = n/2$ for even order.

step	label	note
1. Construction (parts A,B,C,T,U,z; edge rules 1–6)	VALID	Unambiguous. I rebuilt it from the prose alone (<code>build_and_check.py</code>): 37 vertices, 263 edges.
2. Degree table ($z:0$; $u_i:i$; $a_1,a_2,T:13$; $a_3,a_4,a_5:14,15,16$; $c_j:17+j$; $b_j:24+j$)	VALID	Every entry reproduced exactly by code; zero mismatches.
3. $\{d(v)\} = \{0, \dots, 29\}$, $w(G)=30$	VALID	Verified: degree set is exactly $\{0, 1, \dots, 29\}$, $w = 30$.
4. I_1, I_2, I_3 are independent and partition V	VALID	Verified: sizes $21+6+10 = 37$, union is V , all three independent (0 internal edges each). So $\chi \leq 3$.
5. $a_1 b_0 c_0$ is a triangle, hence $\chi = 3$	VALID	Verified; 603 triangles in total. $\chi \geq 3$ re-confirmed two further ways (see Computational check).
6. $\text{RHS} = \lceil 15/7 \rceil = 3$, so $3 > 3$ fails	VALID	$t = 37-30 = 7$, $\text{floor}(30/2)=15$, $\lceil 15/7 \rceil=3$. Counterexample confirmed.
7. Reformulation: conjecture $\Leftrightarrow \text{floor}(w/2) \leq (k-1)t \Leftrightarrow n \leq (2k-1)t+1$ (eq. 1)	VALID	I re-derived it and also checked the equivalence exhaustively over $2 \leq n \leq 199$, all w , all k : 0 disagreements. Graph: $37 > 5 \cdot 7 + 1 = 36$.

step	label	note
8. Reduction of minimality to “bipartite” + “ $t \leq 6$ ”	VALID	A counterexample has $k \geq 3$, so $n \geq (2k-1)t+2 \geq 5t+2$; $n \leq 36$ forces $t \leq 6$. Arithmetic correct.
9. §3.1 bipartite, no isolated vertices: $w \leq b$, $w \leq 2a$, hence $n \leq 3t$ (eq. 2)	VALID	Degrees of the B-side lie in $\{1..a\}$ ($\leq a$ values), the A-side has a vertices, so $w \leq 2a$; all degrees $\leq b$ so $w \leq b$; $n = a+b \geq 3w/2$. Correct, and correct for disconnected bipartite graphs too (any fixed bipartition works).
10. §3.1 with $r \geq 1$ isolated vertices: $n \leq 3t-2r+3 \leq 3t+1$	VALID	$n_0 = n-r$, $w_0 = w-1$, $t_0 = t-r+1$; substitution correct. Edgeless case ($k=1$) satisfies eq. (1) with equality.
11. §3.2 eq. (3): $D = \sum d_i - b = 2e(H) - 2e(L)$, D even	VALID	Both identities re-derived ($\sum_H d = 2e(H) + e(H, L)$).
12. §3.2 eq. (4): $D \geq D_0 = \text{floor}(w^2/4)$	VALID	Min total distance from an integer to w distinct integers is $r(r+1)$ ($w=2r+1$) or r^2 ($w=2r$), both $= \text{floor}(w^2/4)$.
13. §3.2 eqs. (5),(6) refinements using $\delta = \min \text{ degree}$	VALID	I re-derived both algebraically. With $s = b - \delta$, $u = r - s$: $\sum_{j=0}^{\lfloor 2r \rfloor} j-s = r^2 + r + u^2$ gives (5); $\sum_{j=0}^{\lfloor 2r-1 \rfloor} j-s = r^2 + u(u+1)$ gives (6). Exactly the stated right-hand sides.
14. §3.3 eq. (7): $D + 2e + \sum h_i^2 \leq m^2$	VALID	$D + 2e = 2e(H)$ and $e(H) \leq (m^2 - \sum h_i^2)/2$ since colour classes are independent.
15. §3.3 eq. (8): $b \leq m - a + e$	VALID	The m -th vertex lies in L ; it misses the $\geq a$ same-coloured vertices of H , and has $\leq e(L)$ neighbours inside L .
16. $D_0 = m^2 - tm + \text{floor}(t^2/4)$ and eq. (9) $m^2/k \leq Q_k(m) \leq tm - \text{floor}(t^2/4)$	VALID	$w = 2m - t$; $\text{floor}((2m-t)^2/4) = m^2 - mt + \text{floor}(t^2/4)$. Cauchy–Schwarz gives $Q_k(m) \geq m^2/k$.
17. Table (10): $m \leq k$, $2k-1$, $3k-1$, $4k-2$, $5k-2$, $6k-2$ for $t=1..6$	VALID	Proved the general inequalities by hand <i>and</i> checked by brute force that these are the exact integer maxima of (9) for every k in $2..300$ and t in $1..6$: perfect match. The quoted residuals $1+1/k, 1, 1/2, 1/k, 1+1/k, 3+1/k$ at one past the bound are all correct.
18. $X = D - D_0 \geq 0$, $E = \sum h_i^2 - Q_k(m) \geq 0$ even; eq. (11) $X + 2e + E \leq m^2 - D_0 - Q_k(m)$	VALID	E even because $x^2 = x \pmod 2$.
19. Table (12) for $t=1,3,5,6$: $Q_k(m) \leq \lfloor k, \lfloor 9k-5, \lfloor 25k-18, \lfloor 36k-22$; budgets $0,0,2,1$; $a = 1,2,4,5$	VALID	Each entry recomputed: e.g. $t=6$, $m=6k-2$, balanced split $6 - (k-2) - 5^2$ gives $36k-22$, budget $6m-9 - (36k-22)=1$, $a=5$. The balanced minimiser is unique up to permutation, so $E=0$ does pin down a . (Needs $k \geq 2$, which the writeup states.)
20. Boundaries $t=1,3$ (assume $\delta \geq 1$)	VALID	Budget 0 forces $X=e=E=0$; then $b \leq m-a = r$, while $X=0$ plus (5) forces $b = \delta + r \geq r+1$. Contradiction.
21. Boundary $t=5$ (assume $\delta \geq 1$)	VALID	X even since $D_0=r(r+1)$ and D are even. The “ $E \leq 2$ forces $a \geq 3$ ” transfer argument is correct (moving a unit from a part ≥ 6 to a part ≤ 2 drops the square-sum by ≥ 6). Both $X=0$ (giving $b \leq m-3 < \delta + r = m-2$) and $X=2$ (giving $b \leq m-4$, penalty ≥ 4) contradict.
22. Boundary $t=6$ (no δ assumption)	VALID	$r = 6k-5$ odd so $D_0=r^2$ odd, D even, hence X odd and ≥ 1 ; budget 1 forces $X=1$, $e=E=0$, $a=5$, $b \leq r-2$; then (6) with $\delta \geq 0$ gives $D \geq r^2+2$, i.e. $X \geq 2$. Contradiction.
23. §3.4 eq. (13) for even order, then (14) $n \leq (2k-1)t$ for graphs without isolated vertices	VALID	The “add an isolated vertex to an odd-order graph” step keeps t and k and makes the order even; applying (13) gives (14) in both parity cases. Degenerate $k=1$ never has t even at even order, so the $k \geq 2$ restriction in (10) is harmless.
24. §3.4 final reduction for $r \geq 1$ isolated vertices: $n \leq (2k-1)t - (2k-2)r + (2k-1) \leq (2k-1)t+1$	VALID	The last inequality is $2k-2 \leq (2k-2)r$, true for $r \geq 1$. $t_0 = t-r+1 \leq t \leq 6$, and χ is unchanged by deleting isolated vertices.
25. Conclusion “minimum counterexample order is 37”	VALID	Follows from steps 8, 10, 24. Independently corroborated by my mechanisation (below).

step	label	note
26. Scope remark: the isolated vertex is material; the connected version is not refuted	VALID	Verified computationally ($G - z$ satisfies the inequality) and the honesty of the remark is confirmed by my own analysis, which shows the method cannot settle the $\delta \geq 1$ version.

I found **no** step labelled GAP or ERROR.

B.3 Reference check

- **OpenProblemGarden entry** — fetched live. The statement, the “at least two vertices” restriction, the discussion paragraph and the bibliography match `prompt.md` word for word. The entry is still listed as **open** in the OPG vertex-coloring category (last updated 3 March 2013). No comment, update or claimed resolution on the page.
- **Literature search for a resolution** — four searches (counterexample / solved / valency-variety / Melnikov) returned nothing post-1995 on this specific problem, and the Jensen–Toft problem archive at `imada.sdu.dk` has no update page for §4.13/4.14. I found no evidence the result is `ALREADY_KNOWN`, but absence of evidence here is weak: this is a Low-importance, rarely-cited problem.
- **Jensen–Toft p. 90; Dirac 1964; Nettleton 1960; Vizing 1968; Zykov 1968** — none of these could be obtained in full text (the book is not freely available; the Dirac and Nettleton papers are pre-digital). **This does not damage the proof:** the writeup invokes *no* external theorem. Its argument is self-contained, as it explicitly claims (“No computational search or unverified literature result is used in the argument”). The only exposure is the faithfulness of OPG’s transcription of Jensen–Toft’s wording, discussed under Interpretation.
- The Dirac–Nettleton upper bound $\chi \leq n - \lfloor w/2 \rfloor$ is quoted only as context and is not used anywhere. (For the counterexample it reads $3 \leq 37 - 15 = 22$, consistent.)

B.4 Computational check

Scripts: `verification_astra/scripts/melnikovs_valency_variety_problem/` (`build_and_check.py`, `small_cases.py`, `necessary_conditions.py`), run with system `python3` and `networkx 3.6.1`.

1. **The graph** (rebuilt from the prose of §1 only, not from the degree table):

- $|V| = 37$, $|E| = 263$, 603 triangles, components of sizes $36 + 1$.
- Degree sequence: $z:0, u_i:i$ for $i=1..12$, $a_1,a_2:13$, $t_1..t_6:13$, $a_3,a_4,a_5:14,15,16$, $c_0..c_6:17..23$, $b_0..b_5:24..29$. **Zero mismatches** against the writeup’s degree table.
- Degree set = $\{0,1,\dots,29\}$ exactly, so $w(G) = 30$.
- I_1 (21 vertices), I_2 (6), I_3 (10) partition V and each has **0 internal edges**, so $\chi \leq 3$.
- $\chi \geq 3$ confirmed three independent ways: (i) `nx.is_bipartite(G)\=False`;
- (ii) exhaustive backtracking 2-colouring search fails, 3-colouring search succeeds;
- (iii) hand-rolled BFS 2-colouring of each component forces a monochromatic edge ($u_3 b_4$) in the 36-vertex component. So $\chi(G) = 3$ **exactly** — not merely an upper bound.
- $t = 37 - 30 = 7$, $\lfloor w/2 \rfloor = 15$, $\text{RHS} = \lceil 15/7 \rceil = 3$. The conjecture asserts $3 > 3$: **FALSE**. Equivalently $n = 37 > (2k-1)t+1 = 36$.
- $G - z$: $n=36$, degrees $\{1,\dots,29\}$, $w=29$, $t=7$, $\text{RHS} = \lceil 14/7 \rceil = 2 < 3$, connected. The inequality **holds** there — the isolated vertex is indeed essential.

2. **Sensitivity to the placement of the ceiling** (the formula’s fragile point): with $n,w,k = 37,30,3$ the claim fails only for the stated nesting. $\chi > \lceil \lfloor w/2 \rfloor / t \rceil$: $3 > 3$ false (counterexample). $\chi > \lfloor w/2 \rfloor / t$: $3 > 2.143$ true. $\chi > \lceil w/2 \rceil / t$: true. $\chi > \lfloor \lfloor w/2 \rfloor / t \rfloor$: $3 > 2$ true. So the disproof depends on the ceiling being outside the fraction — which is exactly how both `prompt.md` and the live OPG page render it. Worth stating, because a transcription slip here would void the whole thing.

3. **Exhaustive small cases:** all 1251 graphs on 2..7 vertices (`networkx atlas`) satisfy the conjecture. Notably the minimum slack $(2k-1)t+1-n$ is **0 at every order 2..7**, i.e. the conjecture is tight everywhere — consistent with Zykov’s report that Melnikov showed the bound best possible, and an indication that a counterexample was a priori plausible.

4. **Table (10) verified exactly:** for every k in 2..300 and t in 1..6, the true largest integer m with $m^2/k \leq tm - \lfloor t^2/4 \rfloor$ equals the writeup’s claimed k , $2k-1$, $3k-1$, $4k-2$, $5k-2$, $6k-2$. No off-by-one anywhere.

5. **Independent mechanisation of §3** (`necessary_conditions.py`): I encoded eqs. (4)–(8)+(11) as a feasibility test over (x, e, E) with the parity constraints, and computed the largest even order not excluded, for $k = 2..6$ and $t = 1..7$. The output reproduces the writeup’s (13) exactly — e.g. for $k=3$: max even order is 6, 10, 16, 20, 26, 30, 36 for $t = 1..7$, i.e. $(2k-1)t+1$ for odd $t \leq 5$, $(2k-1)t$ for even t (including the $t=6$ boundary exclusion the writeup has to work for), and 36 for $t=7$. With $\delta \geq 1$ imposed the values drop to 4, 10, 14, 20, 24, 30, 36. Two consequences:

- The minimality claim (37) is corroborated by a second, independently written implementation of the same counting scheme.
- **The construction sits exactly on the boundary the method permits:** $G - z$ realises the maximum even order 36 at $k=3$, $t=7$, $\delta \geq 1$. It is not slack; it was engineered against this bound.

6. **The open residue.** Running the same test for $k=3$ and t up to 40 asking whether a counterexample with $\delta \geq 1$ is excluded: it is excluded for $t \leq 9$, but from $t = 10$ onwards it is **not** (e.g. $t=10$ needs $n \geq 52$ and the method permits even orders up to 52; $t=11$ needs 57 and permits 58; the permitted order grows like $\sim 5.45t$ against the required $5t+2$). So the writeup’s technique provably cannot decide the minimum-degree- ≥ 1 version of Melnikov’s problem, and that version is untouched.

B.5 Caveats

1. **The isolated vertex is the whole mechanism.** $G - z$ has $w = 29$ (odd), so $\text{floor}(w/2) = 14 = (k-1)t$ exactly — tight. Adding an isolated vertex bumps w to 30 and $\text{floor}(w/2)$ to 15 while leaving $n - w$, χ unchanged, consuming the free +1 that the floor grants at odd w . Any reader who reads “graph” as “graph without isolated vertices”, or as “connected graph”, is not served by this counterexample.
2. **Primary source not verified.** Jensen–Toft p. 90 could not be read verbatim; the review rests on the live OPG transcription, which does match prompt.md exactly.
3. **Ceiling placement is load-bearing** (see Computational check 2). Any variant nesting of the floor/ceiling makes the graph a non-counterexample.
4. $k=1$ / **edgeless** / $n=1$ **edge cases**: handled correctly and separately by the writeup; $n=1$ is excluded by the problem statement itself (it is the only case with $n = w$, i.e. zero denominator). For $n \geq 2$, $t \geq 1$ always.
5. **Minimality is over all graphs, not over connected graphs**, and says nothing about how far above $37 \Delta \geq 1$ counterexample (if any) would have to sit.
6. **Novelty is unverified.** The writeup says so. I found no published resolution, but this is a Low-importance problem with little literature, so the search is weak evidence. A 1968-era reader with a tight example in hand could have made this same observation; I cannot rule out that it is folklore.
7. **Gap between the verdict block and what is proved:** the `one_line` says “making the proposed right-hand side equal to 3; moreover, 37 vertices is minimum” — both verified. The `verdict: disproved` is correct for the problem as stated. It would be misleading only if read as “Melnikov’s question is settled”, which the body of the writeup explicitly disclaims.

B.6 Referee summary

I set out to break this and could not. Reconstructing the 37-vertex graph from the prose alone (not from the degree table) reproduces every claimed property exactly: 263 edges, degree set precisely $\{0, 1, \dots, 29\}$ so $w = 30$, a verified partition into three independent sets, and $\chi = 3$ established as an exact value — non-2-colourability confirmed by three independent methods, not just an upper bound from a colouring. With $n - w = 7$ the proposed right-hand side is $\text{ceil}(15/7) = 3$, so the assertion $\chi > 3$ fails, and the live OpenProblemGarden entry states the problem with no hypothesis beyond $|V| \geq 2$ (which is exactly the condition making the denominator nonzero), so this is a genuine counterexample to the problem as the source states it, not to a strawman. The minimality argument — the long part — also survives a full line-by-line audit: I re-derived eqs. (1)–(14) by hand, checked table (10) against brute force for all k up to 300, and re-implemented the whole counting scheme independently, which reproduces the writeup’s bounds exactly and confirms that no counterexample exists below 37 vertices. The one substantive reservation is interpretive rather than mathematical: the counterexample lives entirely on the isolated vertex (deleting it yields a connected 36-vertex graph that satisfies the inequality), so what is refuted is the degenerate-inclusive reading, while the minimum-degree- ≥ 1 question — which is what most readers would call Melnikov’s problem — is untouched; my mechanisation further shows the writeup’s own method cannot decide that version for $t \geq 10$. The writeup states this limitation itself, which is why I do not score it as interpretation gaming. Verdict CONFIRMED, with the caveat that this is a correct disproof of the literal statement rather than a resolution of the underlying question, and that the primary source (Jensen–Toft p. 90) could not be read verbatim.

References

- [1] Open Problem Garden, *Melnikov’s valency-variety problem*, posted 3 March 2013, attributed to L. S. Melnikov, http://www.openproblemgarden.org/op/melnikovs_valency_variety_problem. Also catalogued at https://graph-theory-ai.github.io/graph-conjectures/op/melnikovs_valency_variety_problem/.
- [2] T. R. Jensen and B. Toft, *Graph Coloring Problems*, Wiley-Interscience Series in Discrete Mathematics and Optimization, John Wiley & Sons, New York, 1995; p. 90.
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- [6] A. A. Zykov, Problem 11, in: H. Sachs, H.-J. Voss and H. Walther (eds.), *Beiträge zur Graphentheorie vorgetragen auf dem Internationalen Kolloquium in Manebach DDR vom 9.–12. Mai 1967*, B. G. Teubner, Leipzig, 1968, p. 228.